

Database Project – SkyTrack Airline System

Database description:

The SkyTrack Airline System is designed to manage the core operations of a regional airline, including airports, aircraft, flights, passengers, bookings, and crew assignments. Each Airport is identified by a unique IATA code, name, city, and country. Each Aircraft has a registration number, model, manufacturer, total seating capacity, and year of manufacture. A Flight has a unique flight number, departure datetime, arrival datetime, and status. Every flight must be assigned exactly one aircraft, and it departs from one origin airport and arrives at one destination airport — these are two separate relationships between Flight and Airport. A Passenger holds a unique national ID, full name, email, phone, nationality, and date of birth. A Booking links one passenger to one flight and records the seat number, class, price paid, and booking date. A Crew Member has a full name, role, and a unique license number. One crew member can serve on many flights and one flight can have many crew members, forming a many-to-many relationship that is managed through a FlightCrew association table.

1- Design a complete ERD diagram of the above system that includes:

- Entities and their attributes
- Primary keys (underlined or clearly marked)
- All relationships with their names
- Cardinality for each relationship (1:1, 1:M, M:N)
- Participation constraints (total or partial) for each entity in every relationship

2- Map the ERD diagram into a logical schema applying the conversion rules, and also apply the normalization rules (First Normal Form, Second Normal Form, Third Normal Form) showing the final tables structure along with the referential relationships.

3- Implement the physical schema of this database by converting the logical schema into SQL code that creates the above database, and apply the following constraints:

– Identity and Keys:

- All IDs should be set as IDENTITY.
- Foreign key columns should NOT be IDENTITY.
- All foreign keys should include ON DELETE CASCADE and ON UPDATE CASCADE.

– The following must be UNIQUE:

- Flight number
- Aircraft registration number
- Passenger email
- Passenger national ID
- Crew member license number

– The following fields must be NOT NULL:

- Airport: IATA code, Name, City, Country

- Aircraft: Registration number, Model, Manufacturer, Capacity
- Flight: Flight number, Departure datetime, Arrival datetime, Status
- Passenger: Full name, National ID, Email, Nationality, Date of birth
- Booking: Seat number, Class, Price, Booking date
- Crew Member: Full name, Role, License number

– **Use CHECK constraints:**

- Flight Status allows only: 'Scheduled', 'Delayed', 'Cancelled', 'Completed'
- Booking Class allows only: 'Economy', 'Business', 'First'
- Aircraft Capacity must be greater than zero
- Booking Price must be greater than zero
- Arrival datetime must be greater than Departure datetime
- Crew Member Role allows only: 'Pilot', 'Co-Pilot', 'Flight Attendant', 'Engineer'

– **Use DEFAULT values:**

- Flight Status defaults to 'Scheduled'
- Booking date defaults to the current date

SkyTrack Airline System – SQL Practice

Task Objective

In this section you will work directly with the database you created. You will populate it with data, make modifications, and then write a series of queries to extract meaningful information from the system. Complete each part in order before moving to the next.

Part 1: Insert Sample Data

Populate your database with realistic sample data that covers all tables. Your inserted data must satisfy all constraints defined in your schema and must be sufficient to produce meaningful results when queried in the next section.

Minimum insertion requirements:

- At least 5 airports from different countries
- At least 5 aircraft with different models and manufacturers
- At least 8 flights — cover all four status values: Scheduled, Delayed, Cancelled, and Completed. Flights must use different origin and destination airports and different aircraft.
- At least 8 passengers from different nationalities
- At least 10 bookings distributed across different flights and passengers. Include all three booking classes: Economy, Business, and First.
- At least 6 crew members covering all four roles: Pilot, Co-Pilot, Flight Attendant, and Engineer
- Assign crew to all flights through the FlightCrew table. Each flight must have at least one pilot and at least one flight attendant.

Part 2: UPDATE and DELETE

UPDATE Tasks

1. Update one flight status from 'Scheduled' to 'Completed'.
2. Change one flight status from 'Delayed' to 'Cancelled'.
3. Increase all Economy class booking prices by 10%.
4. Update one passenger's phone number.
5. Move one crew member to a different role.

DELETE Tasks

Before each DELETE, run a SELECT first to confirm the row exists. Include both the SELECT and the DELETE in your submission.

1. Delete one cancelled flight.
2. Delete one booking linked to a cancelled flight.
3. Try to delete a passenger who has existing bookings. Observe what happens and write a short comment in your SQL file explaining the result.

SkyTrack Airline System – Query Practice

Part 3: Data Queries

Write SQL SELECT statements to answer each question below. Read every question carefully — some answers come from a single table, others require you to look across multiple tables. Choose the right approach for each one.

Basic Level

1. List all flights and their current status, ordered by departure datetime from earliest to latest.
2. Show all passengers, ordered alphabetically by full name.
3. List all aircraft and their seating capacity, ordered from largest to smallest.
4. Show all bookings and their class. Display only distinct class values that exist in the system.
5. List all flights that have a status of 'Delayed' or 'Cancelled'.
6. Show all passengers whose nationality is 'Omani'.
7. List all airports, ordered by country.

Medium Level

1. For each flight, show the flight number, the name of the origin airport, and the name of the destination airport.
2. Show each booking along with the full name of the passenger who made it and the flight number it belongs to.
3. List all crew members assigned to flight 'SK101', showing their full name and role.
4. Show all completed flights along with the aircraft model used on each flight.
5. For each passenger, show their full name and the total number of bookings they have made. Order by booking count from highest to lowest.
6. Show the total revenue collected from each booking class.
7. Count how many flights each aircraft has been assigned to.
8. List all flights that have more than one booking.
9. Show the full details of all bookings — passenger name, flight number, origin airport, destination airport, class, and price paid.

Advanced Level

1. Show each flight with its flight number, origin airport, destination airport, aircraft model, and the total number of passengers booked on it. Include flights that have no bookings.
2. List all passengers who have never made a booking.
3. For each flight, show the flight number and the total revenue generated from its bookings. Show only flights where the total revenue exceeds 500. Order from highest to lowest.
4. Show each crew member's full name and the total number of flights they have been assigned to. Show only crew members assigned to more than one flight.

5. Find the average booking price per flight. Show only flights where the average price is above the overall average price across all bookings.
6. Show the flight with the highest number of bookings. Display its flight number, origin, destination, and total bookings.
7. For each booking class, show the total revenue, the number of bookings, the average price, the highest price, and the lowest price.
8. List all passengers who booked a flight that is currently 'Cancelled'. Show the passenger name, flight number, and booking date.
9. Show all flights that have at least one pilot and at least one flight attendant assigned. Display the flight number, total crew count, and departure datetime.
10. FINAL CHALLENGE: Show the complete flight summary — flight number, origin airport city, destination airport city, aircraft model, aircraft manufacturer, total passengers booked, total crew assigned, and total revenue. Order by total revenue from highest to lowest.

BONUS TASK – Indexing

This is a self-research task. No examples, hints, or code are provided. You are expected to independently research the concept of database indexing, understand how it works, apply it to the SkyTrack Airline database, and justify every decision you make in writing.

PART 1: RESEARCH

Focus: understand the concept before you write a single line of SQL

1. What is a database index and what problem does it solve? Explain in your own words.
2. What is the difference between a Clustered Index and a Non-Clustered Index? How many clustered indexes can one table have, and why?
3. What is a Unique Index? Identify at least two columns in the SkyTrack database where a unique index is naturally suitable.
4. What is a Composite Index? Describe a situation in the airline system where combining two columns in one index would improve a query.
5. What trade-off does adding an index introduce? Think carefully about INSERT, UPDATE, and DELETE operations.

PART 2: IMPLEMENTATION

Apply only where it makes sense — justify every choice

1. Review the queries you wrote in the query practice section. Identify columns that appear frequently in WHERE conditions, JOIN conditions, or ORDER BY clauses. Create at least 4 indexes across the SkyTrack Airline database based on this analysis.
2. For each index you create, add a comment block directly above it in your SQL file that states:
– The table and column(s) the index is on – The type of index chosen (clustered, non-clustered, unique, composite) – Which query from your submission benefits from it – Why this column was selected (e.g. high cardinality, frequent filter, join key)
3. Identify at least one column where creating an index would NOT be beneficial or could hurt performance. Write a comment in your SQL file explaining your reasoning.
4. In your README, answer the following: if the SkyTrack system processes thousands of bookings and flight status updates every day, which two of your indexes would you consider the most critical? Justify your answer.

RULES

- Do NOT copy index examples from the internet without understanding them
- Every index must be justified with a comment in your SQL file
- Each index must be linked to a real query from your project
- Include all index scripts inside the BONUS/ folder in your repository

PURPOSE

Today you learn:

- Queries can be fast or slow — indexes make the difference
 - Design decisions have a cost and a benefit
 - A good DBA thinks before adding indexes
 - Performance is part of database design, not an afterthought
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Submission

Upload your entire project as a single public GitHub repository. The repository must be named exactly:
`database-project`

Repository structure:

```
database-project/  
  ■■■ 01-ERD/  
  ■■■ 02-Mapping/  
  ■■■ 03-Create-Tables.sql  
  ■■■ 04-Insert-Data.sql  
  ■■■ 05-Update-Delete.sql  
  ■■■ 06-Queries-Basic.sql  
  ■■■ 07-Queries-Medium.sql  
  ■■■ 08-Queries-Advanced.sql  
  ■■■ BONUS/  
  ■ ■■■ indexing.sql  
  ■■■ README.md
```

README must include:

- A brief description of the SkyTrack Airline System
- ERD summary — entities, key relationships, and any design decisions
- Mapping decisions — where you placed foreign keys and why
- Any errors you faced during INSERT or DELETE and how you resolved them
- The difference between WHERE and HAVING in your own words
- The most useful query you wrote and why
- BONUS: which indexes you added and the full justification for each